

CCNP CORE & ADVANCE



CORE CCNP

Course Overview & Exam Objectives

- Introduction to the CCNP Core (350-401 ENCOR) exam
- Review of course objectives, structure, and exam topics

Network Architecture & Design

- Enterprise network design principles and best practices
- Scalable architectures, redundancy, and high availability
- Designing resilient networks with appropriate topologies

Virtualization & Software-Defined Networking (SDN)

- Overview of network virtualization technologies
- Concepts of SDN and Network Function Virtualization (NFV)
- Controller-based networking and centralized management

Infrastructure Technologies

- Layer 2 and Layer 3 switching and routing fundamentals
- In-depth study of routing protocols (OSPF, EIGRP, BGP)
- VLANs, STP, inter-VLAN routing, and network segmentation

Network Assurance & Troubleshooting

- Network monitoring and performance management
- Troubleshooting methodologies and diagnostic tools
- Quality of Service (QoS) principles and implementation

Network Security

- Fundamentals of secure network design
- Access control, segmentation, and threat mitigation strategies
- Implementation of security policies and firewall integration

Automation & Programmability

- Introduction to network automation tools and scripting (Python)
- REST APIs, NETCONF, and automation workflows
- Implementing configuration management and controller-based automation

Hands-on Labs & Exam Preparation

- Practical lab exercises simulating real-world scenarios
- Practice tests and exam preparation sessions
- Review sessions and case studies to reinforce learning

Advance CCNP

Advanced Routing Protocols

- Deep dive into BGP optimization, advanced OSPF configurations, and route filtering
- MPLS VPN and inter-domain routing strategies
- Policy-based routing and traffic engineering

Enhanced Switching & Layer 2 Technologies

- Advanced VLAN design, multicast, and redundancy protocols
- Enhanced STP/RSTP/MSTP configurations and troubleshooting
- Virtual switching and software-defined networking principles

WAN & Data Center Technologies

- SD-WAN, DMVPN, and IPsec VPN deep dives
- Data center networking architectures and advanced MPLS implementations
- Interconnectivity solutions and high-availability design

Network Security & Automation

- Advanced firewall configurations, ACLs, and segmentation strategies
- Integration of network security into routing/switching
- Automation with Python, REST API integration, and SDN deployments

Advanced Troubleshooting & Performance Optimization

- Proactive monitoring, diagnostic tools, and simulation labs
- Troubleshooting complex network scenarios and performance tuning
- Real-world case studies and lab simulations

Advanced Network Architecture & Design

- Designing scalable, resilient enterprise networks
- Best practices for high availability and disaster recovery
- End-to-end network design and implementation strategies

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